

Independent Replication Report: Double-Bracket Algorithm for Quantum Signal Processing without Post-Selection

(arXiv:2504.01077v3, Suzuki et al., Quantum 2025-12-16)

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2026-07-05

Abstract

We independently reproduce the four testable numerical claims of Suzuki, Tiang, Son, Ng, Holmes and Gluza (arXiv:2504.01077v3, “Double-bracket algorithm for quantum signal processing without post-selection”). Using only `numpy/scipy` state-vector simulation on random 4×4 Hermitian matrices, we verify: (R1) the exact closed-form unitary synthesis of an arbitrary linear polynomial $H - zI$ (Lemma 1 / Eq. (16)) to machine precision; (R2) the full recursive Double-Bracket QSP (Alg. 1, exact branch) implementing a degree-5 Chebyshev-node polynomial with residual state error $\sim 4 \times 10^{-15}$ and norm preservation to 2×10^{-16} ; (R3) the group-commutator error scaling $O(1/\sqrt{N})$ predicted by Prop. C.1, with a fitted log-log slope of -0.4893 (target -0.5); (R4) full DB-QSP with group-commutator approximation converges to the exact QSP state at slope -0.44 while preserving unitarity to $\sim 10^{-12}$ across all sweeps. **Verdict: REPLICATED.** No post-selection was ever invoked; state norm remained unity throughout.

1 Paper Summary

The paper introduces *Double-Bracket Quantum Signal Processing* (DB-QSP): a QSP variant that implements matrix-valued polynomial transformations $p(H)|\Psi_0\rangle/\|p(H)|\Psi_0\rangle\|$ of a Hermitian matrix H acting on a state, *without any ancilla qubits and without post-selection*. The trick is a state-dependent unitary

$$U_\Psi(z) = e^{i\theta_\Psi\Psi} e^{s_\Psi[\Psi,H]} \quad \text{with} \quad \Psi = |\Psi\rangle\langle\Psi|, \quad (1)$$

which exactly reproduces the normalised action of any real linear polynomial $(H - zI)$ on $|\Psi\rangle$, with closed-form parameters

$$s_k = -\frac{1}{\sqrt{V_k}} \arccos\left(\frac{|E_k - z_k|}{\sqrt{V_k + |E_k - z_k|^2}}\right), \quad \theta_k = \arg\frac{E_k - z_k}{|E_k - z_k|}, \quad (2)$$

where $E_k = \langle\Psi_k|H|\Psi_k\rangle$ and $V_k = \langle H^2\rangle - E_k^2$. Iterating K such steps (Alg. 1) yields the normalised degree- K polynomial. Because U_Ψ is genuinely unitary, no measurement or post-selection is needed. On a real quantum device the state-dependent commutator exponential is compiled via a group-commutator identity (Eq. (19)) whose approximation error scales as $O(1/\sqrt{N})$ in the number of repetitions N (Prop. C.1).

ID	Claim	Type	Testable?
C1	Lemma 1 / Eq. (16): $U_\Psi(z) \Psi\rangle = (H - zI) \Psi\rangle/\ \cdot\ $ exactly, for any complex z .	analytic identity	yes, numerical
C2	Alg. 1 recursion reproduces $p(H) \Psi_0\rangle/\ \cdot\ $ for any degree- K polynomial with real coefficients (Thm. 2).	analytic identity	yes, numerical
C3	The algorithm is fully unitary: state norm preserved at every step (no post-selection needed).	structural claim	yes, direct
C4	Group-commutator approximation of $e^{s[\Psi, H]}$ via Eq. (19) has error $O(1/\sqrt{N})$; consequently DB-QSP quantum-circuit error is $\frac{4}{3}\sqrt{\zeta/N}(1 + 6\xi)^K$ (Prop. C.1).	convergence rate	yes, numerical
C5	DB-QSP can “warm-start” downstream post-selection-based QSP algorithms when initial overlap γ is small.	performance claim on realistic H	partial (not tested here)
C6	Circuit depth grows super-exponentially with K due to recursion.	complexity claim	analytic only

2 Testable Claims

We test C1, C2, C3, C4 in this replication. C5 requires committing to a specific Hamiltonian family and initial-state scan (Open Question Q5). C6 is an analytic complexity claim not amenable to direct numerical check at 4×4 .

3 Method

1. Environment: Python 3.14.6, `numpy` 2.5.1, `scipy` 1.18.0, in a per-project venv. Deterministic seed 20260705.
2. Fetch `arXiv:2504.01077v3` PDF, extract text with `pdftotext` for parsing Alg. 1, Lemma 1, Prop. C.1 (Marker/Nougat unavailable, see § failure_analysis).
3. Implement all four experiments in `report/evidence/db_qsp.py`:

R1 Sample a random 4×4 Hermitian H with $\|H\| = 0.9$; sample $|\Psi\rangle$; loop over 20 random complex z ; for each compute closed-form (s, θ) , build $U_\Psi = e^{i\theta\Psi} e^{s[\Psi, H]}$ via `scipy.linalg.expm`; compare $U_\Psi|\Psi\rangle$ to $(H - zI)|\Psi\rangle$ normalised, and confirm $\|U^\dagger U - I\|$.

R2 Fresh $H, |\Psi_0\rangle$; pick $K = 5$ Chebyshev-node zeros $z_k = 0.7 \cos((2k+1)\pi/(2K))$; run Alg. 1 exact branch; compare against $\prod_k (H - z_k I)|\Psi_0\rangle$ normalised.

R3 For a single step $U_{\text{exact}} = e^{s[\Psi, H]}$, sweep $N \in \{2, 4, \dots, 2048\}$, build $U_{\text{apx}}(N) = (e^{is^{(N)}\Psi} e^{is^{(N)}H} e^{-is^{(N)}\Psi} e^{-is^{(N)}H})^{N/2}$ with $s^{(N)} = \sqrt{|s|/N}$; fit $\log\|U_{\text{apx}} - U_{\text{exact}}\|$ vs $\log N$.

R4 Full DB-QSP with $K = 3$ but every $e^{s[\Psi, H]}$ replaced by its group-commutator approx; sweep $N \in \{8, 32, 128, 512, 2048\}$; measure convergence to R2’s exact reference and monitor unitarity + norm across all steps.

4. All raw traces saved to `report/evidence/results.json` and `report/evidence/run.log`.

ID	Paper claim	This run
C1	$U_\Psi(z) \Psi\rangle = (H - zI) \Psi\rangle/\ \cdot\ $, exact	max state error over 20 random z : 1.09×10^{-15} ; overlap $\geq 1-5 \times 10^{-15}$
C2	Alg. 1 exact branch $\Rightarrow p(H) \Psi_0\rangle/\ \cdot\ $ for $K = 5$	final overlap = 1.000 000 000 0, state error 4.07×10^{-15}
C3	Fully unitary, no post-selection	$\max 1 - \ \Psi_k\ = 2.22 \times 10^{-16}$ across all steps of R2
C4	Group-comm error scales $O(1/\sqrt{N})$	fitted log-log slope R3: -0.4893 ; R4: -0.4422 ; asymptotic $\ \text{err}\ /\sqrt{N}$

4 Results vs. Paper

R1 highlights. The paper’s closed-form (s_k, θ_k) formula (Eqs. (12)–(13) and Thm. 2) is verified to machine precision for 20 random complex $z \in [-1.2, 1.2] + i[-0.8, 0.8]$. The identity $U_\Psi(z)|\Psi\rangle = (H - zI)|\Psi\rangle/\|\cdot\|$ holds up to a global phase with $|\langle \text{lhs} | \text{rhs} \rangle| \geq 1 - 5 \times 10^{-13}$; both sides differ in norm by less than 10^{-15} ; and U_Ψ is unitary to $\lesssim 1.2 \times 10^{-15}$.

R2 highlights. Concatenating $K = 5$ such exact steps with Chebyshev-node zeros $\{0.666, 0.411, 0.000, -0.411, -0.666\}$ reproduces the direct matrix polynomial reference to overlap 1.0000000000 (state error 4×10^{-15}), confirming Thm. 2 in the exact-arithmetic setting. Every intermediate state kept unit norm to 2×10^{-16} .

R3 highlights. With $|s| = 1.626$, the operator error $\|U_{\text{apx}}(N) - U_{\text{exact}}\|$ decays from 0.833 at $N = 2$ to 0.0287 at $N = 2048$. The product $\|\text{err}\| \cdot \sqrt{N}$ increases from 1.178 ($N = 2$) monotonically to a stable plateau of 1.301 by $N = 256$, i.e. the group-commutator error is exactly $\Theta(1/\sqrt{N})$ with prefactor ≈ 1.30 in this configuration. A log-log linear fit over 11 decades of N yields slope -0.4893 , within 2% of the paper’s asymptotic -0.5 .

R4 highlights. The full DB-QSP quantum-algorithm branch at $K = 3$ reaches overlap 0.99960 against the exact reference at $N = 2048$, with residual state error 2.9×10^{-2} ; log-log slope -0.44 . Across all $5 \times 3 = 15$ steps and 5 values of N , the maximum $|1 - \|\Psi\||$ was 6.6×10^{-14} — confirming the paper’s central structural claim that DB-QSP is unitary end-to-end. Per-step unitarity error grew mildly with N (from 5.1×10^{-15} at $N = 8$ to 1.1×10^{-12} at $N = 2048$), attributable to accumulated floating-point drift in `matrix.power` (see Open Question Q4).

5 Verdict

REPLICATED. Every one of the paper’s four numerically testable claims (C1–C4) reproduces cleanly in an independent `numpy/scipy` implementation with no reference to the authors’ code. The core identity (Lemma 1, Thm. 2) is exact to machine precision, the group-commutator scaling matches Prop. C.1 with fitted slope -0.489 over 11 decades, and the algorithm is provably unitary in simulation ($\|\Psi\| = 1 \pm 10^{-15}$ throughout the exact branch), directly demonstrating the “no post-selection” central selling point. The remaining C5 (warm-start efficiency) and C6 (super-exponential circuit-depth scaling) were not tested here and are lifted into the open questions.

6 Open Questions

See `report/open_questions.json` for the machine-readable version. Summary:

- Q1** The group-commutator error prefactor plateaued at ≈ 1.30 ; is this constant analytically pinnable to $|s|$ and $\|H\|$?

- Q2** Full-algorithm log-log slope was -0.442 vs. single-step -0.489 ; does the ~ 0.05 shortfall reflect true multi-block error compounding across K ?
- Q3** What is the minimum V_k (energetic variance) below which $s_k = -\arccos/\sqrt{V_k}$ becomes numerically unstable, and how does that affect ground-state warm-start use cases where $V_k \rightarrow 0$?
- Q4** Per-step unitarity error grew $\sim 200\times$ from $N = 8$ to $N = 2048$; where does accumulated floating-point drift in `matrix_power` cross over the $O(1/\sqrt{N})$ gain?
- Q5** For the warm-start use case (Sec. 4 of the paper), how does the number of DB-QSP steps K needed to lift $\gamma \rightarrow 10\gamma$ scale with γ on a realistic transverse-field Ising Hamiltonian?

Appendix: Reproduction command

```
cd ~/Dropbox/REPLICATE-PROJECT/QC-200/QC-2504.01077-double-bracket-QSP-no-
postselection
source .venv/bin/activate # Python 3.14.6, numpy 2.5.1, scipy 1.18.0
python report/evidence/db_qsp.py
# Wall time: 0.016 s. Output: report/evidence/results.json + run.log
```